

Rockridge Power & Light Company

Meter Testing Program Plan

RPL-MTR-PGM-001 · Version 6.0

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Owner	Gregory Walsh, Manager, Meter Shop & Standards Laboratory
Reviewer	Angela Ruiz, Director, Metering
Approver	Michael Brennan, Vice President, Operations
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1. Purpose

This Meter Testing Program Plan describes how Rockridge Power & Light Company (RPL) ensures the accuracy, integrity, and regulatory compliance of its meter population. The plan governs new-meter acceptance testing, in-service testing (periodic and statistical sampling), test equipment calibration and traceability, customer-requested testing, and annual program reporting. It applies to all 409,950 revenue meters in RPL's service territory as of 2024-12-31.

2. Scope

This plan covers the full RPL revenue meter fleet: 371,200 solid-state AMI meters, 31,400 solid-state AMR meters, and 7,350 electromechanical meters (total: 409,950) within RPL's 14-county service territory in west-central Indiana.

3. Definitions

As-Found Accuracy: Measured percentage registration before any adjustment. (170 IAC 4-1-8(a))

Average Accuracy: $(FL + LL) \div 2$. (170 IAC 4-1-8(a))

Full Load (FL): 100% of rated test amperes. **Light Load (LL):** 10% of rated test amperes.

AQL: 2.50 (170 IAC 4-1-10(c)(4)). **U/L:** 102%/98% (170 IAC 4-1-10(c)(5)).

Method A: 16-year periodic. **Method B:** Annual statistical sampling.

4. Regulatory Basis

Governing regulation: 170 IAC 4-1 (Indiana Administrative Code), law as of 2024-12-31. ANSI C12.1, ANSI C12.20, and ANSI/ASQC Z1.9-1993 referenced as company practice.

5. Organization and Roles

Gregory Walsh (P19) — Manager, Meter Shop & Standards Laboratory: day-to-day program management.

Angela Ruiz (P20) — Director, Metering: program ownership, IURC interface.

Luis Hernandez (P18) — Supervisor, Metering Services: field testing coordination.

Michael Brennan (P16) — VP Operations: approver.

6. Meter Records

Test records must include: meter ID, reason for test, pre-test reading, and all accuracy data. (170 IAC 4-1-4(a)) Records are maintained in WAM. Minimum retention: 3 years. (170 IAC 4-1-3)

7. Meter Location and Accessibility

Outdoor installation preferred. (170 IAC 4-1-5(A)) Mounting height: 4–6 ft above standing surface. (170 IAC 4-1-5(B),(C)) Minimum board spacing: 7.5 in. (170 IAC 4-1-5(C))

8. New and Repaired Meter Acceptance

New non-self-contained meters: inspected, tested, adjusted before installation. (170 IAC 4-1-6(a)) RPL also performs acceptance sampling on each lot per §9.3.

Non-self-contained meters: tested prior to installation or within 60 days. (170 IAC 4-1-6(c),(d))

9. In-Service Testing Program

53 homogeneous groups; minimum group size 301 meters. (170 IAC 4-1-10(c)(2))

Method B for all solid-state groups; Method A 16-year periodic for EM meters.

Sample size table (company practice, consistent with ANSI/ASQC Z1.9-1993 Inspection Level II).

Lot acceptance: 2.50 AQL, U=102%, L=98% DSL method. (170 IAC 4-1-10(c)(4),(5))

Rejected lot accelerated testing: maximum 96 months. (170 IAC 4-1-10(c)(7))

10. Accuracy Limits and Average-Accuracy Method

Average: $\leq \pm 2.00\%$; FL: $\leq \pm 1.00\%$; LL: $\leq \pm 3.00\%$. (170 IAC 4-1-9(b)(1))

PF test (CT meters): 100% rated current, 50% lagging PF, $\leq \pm 2.00\%$. (170 IAC 4-1-9(b)(5))

No-load prohibition: < 1 revolution at no-load, voltage < 110% standard. (170 IAC 4-1-9(a))

11. Test Equipment and Standards

Traceability: NIST → Accredited Lab → Reference Standards (3) → Test Boards (8) → Portables (32).

Reference standard recal interval: ≤ 2 years. (170 IAC 4-1-7(C))

Portable standard out-of-service if error $> \pm 1\%$. (170 IAC 4-1-7(D))

12. AMI Meter Health Monitoring

Company practice supplementing Method B. AMI HES flags: zero-consumption, tamper, power quality, metrology diagnostic, disconnect switch failure. Removal-for-test after 14 consecutive days.

13. Customer-Requested Tests

First and second tests in any 12-month period are free. (170 IAC 4-1-11(a))

Written report within 10 days. (170 IAC 4-1-11(d)) Customer appeal within 5 days. (170 IAC 4-1-11(e))

Fee (where applicable): \$40.00 residential/single-phase; \$95.00 polyphase/demand.

14. 2024 Program Results

Total tests: 5,724. Within-limits rates: AMI 99.73%; AMR 99.76%; EM 95.33%. No group acceptance failure in 2024. Customer tests: 620.

15. 2025 Test Plan

Annual sample draw executed by 2025-01-15. 3,750 meters selected across 53 groups (data/inservice_sample_selection_2025.csv). Method A EM periodic tests scheduled in WAM.

16–21. KPIs, Records, Training, Related Documents, Revision History, Approval

KPIs: $\geq 98\%$ annual sample completion; 100% customer test report compliance; 100% reference standard recal on schedule. Records retention: 3-year minimum. (170 IAC 4-1-3)

Revision: v6.0 (2025-01-08) — updated for 2024-12-31 law snapshot.

21. Approval Block

Role	Name / Title	Date
Owner	Gregory Walsh Manager, Meter Shop & Standards Laboratory	2025-01-08
Reviewer	Angela Ruiz Director, Metering	2025-01-08

Approver	Michael Brennan Vice President, Operations	2025-01-08
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